Cristóbal Middleton Frederick

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PROFESSIONAL PROFILE

Industrial engineer focused on turning complex business problems into data-driven solutions. Experienced in putting predictive and forecasting models into production, automating analytical processes and demand planning, as a **data scientist** at **Macal** and demand planner at **Puig**. I created and run two ventures of my own: **Tordo**, an open site with Chile's public procurement data, and **PixelFrame**, an online store for mosaic wall art. Driven by challenges, innovation and projects with measurable impact, alongside technical and business teams.

WORK EXPERIENCE

Puig Santiago, Chile
Demand Planner Executive Aug 2026 – Nov 2026

- Managed **demand planning** for a portfolio of $\sim$1,700 SKUs (Rabanne, Carolina Herrera Makeup, Banderas, Agatha Ruiz de la Prada and Quorum, among others), forecasting demand for each SKU.
- Built a **demand database in DuckDB** integrating the SAP IBP forecast, SAP BW sales history and customer sell-out, with an **automated, validated daily load**.
- Developed a **demand forecasting model** that blends a seasonal baseline with the bias-corrected IBP forecast: **+11 pts Forecast Accuracy** over IBP in backtesting, with lower bias. Delivered through a **Streamlit** application.

Macal Santiago, Chile
Data Scientist Aug 2024 – Aug 2026

- Developed an **ensemble model to predict apartment prices** with automated retraining and **deployment** via **GCP endpoints (Cloud Run connected to GitHub)**. Achieved **17% lower error** than the previous method and **>1000x time reduction** (**+18 hrs/week** saved).
- Developed **forecasting models**: vehicle sales amount projection (**23% lower error**) and an **optimal investment budget** estimation model in collaboration with the marketing team.
- Developed a **text matching and normalization algorithm** (brand-model-version) for the automotive industry, integrating **API-based data fallback**.
- **MLOps**: End-to-end development, **productionization**, monitoring, and maintenance of predictive models.
- **Automation** and **orchestration of pipelines** (ETL/ELT) hosted on departmental servers.
- Built **QuickSight and Looker dashboards** and reports with **KPIs** for tracking.

Working Holiday Visa Sydney/Lamington, Australia
Potts Point Hotel & Lamington National Park Jul 2023 – Jun 2024

- One-year international experience working in **high-pressure environments** requiring rapid cultural adaptation.
- Strengthened **interpersonal skills**, **problem-solving** abilities, and **self-management**.

Macal Santiago, Chile
Product Manager BI Jul 2021 – Apr 2023

- End-to-end management of vehicle product lines and complementary services. Generated **actionable insights** for business growth.
- Created a **financial product** that captured **91% of previously unaddressed traffic**, increasing **profitability by 12%** compared to prior credit products.
- Implemented **pricing improvements** based on demand and vehicle/customer segmentation, resulting in a **35% increase** in bookings within the most profitable segment.
- Led continuous development of **fintech and insurance products** in coordination with lenders, UX, UI, PO, and IT teams.
- Developed **Tableau dashboards**, reports, and **KPIs** for performance monitoring.

Spurgeon Santiago, Chile
Project Manager (side project) 2022 (5 months)

- Managed the build and sale of a **software product for Lipigas**.

Los Parques S.A. Santiago, Chile
Management Advisor Oct 2020 – May 2021

- Developed and managed **KPIs** for monitoring sales force performance and targets.
- **Automated reporting** and provided support for **data governance** initiatives.

- Supervised the design and development of a **mobile application** for client portfolio management focused on digital transformation. Trained **200+ executives**. Reduced management-contact time by **90%** and increased contacted clients by **20%**.
- Designed marketing campaigns, conducted market analysis, and built the **annual budget**.
- Supervised the **commercial call center**. Led operational reorganization resulting in a **1.3x increase in occupancy** and a **+15% sales growth**.

Los Parques S.A. Santiago, Chile
Business Intelligence Analyst Jun 2019 – Sep 2020

- Portfolio analysis, campaign redesign, and construction of **segmented commercial strategies**. Achieved a sustained **+3% monthly increase** in campaign performance.
- Developed **lead assignment flows** and campaign automation, reducing creation and distribution time by **15x**.
- Provided support in **management control**, market research, and commercial reporting.

Mercado Libre Santiago, Chile
Intern Aug 2018 – Jan 2019

- Analyzed **e-commerce logistics operations** and explored **data-driven strategies**.
- Developed a **logistics provider evaluation project** and managed buyer response **SLAs**.

PROJECTS

Tordo – tordo.cl Oct 2025 – Present

- Free public site with Chile's **public procurement data** (Mercado Público) for small businesses that sell to the State: prices paid per product, competition per market and today's open quick purchases. In production since October 2026.
- **Polars** pipeline (hourly ingestion, weekly analytics) over three years of purchase orders, materialized in **PostgreSQL** and served with **FastAPI**, **Next.js** and an **MCP** server, with **CI and 2,000+ tests**. Open source: github.com/cris-middleton/tordo.
- Published **data notes** on errors found and how they were fixed; e.g. orders in UF that inflated municipal spending to CLP 14 trillion a year, against the official 4.15.

PixelFrame – pixelframe.cl Apr 2026 – Present

- Online store for mosaic wall art made with compatible building blocks, built with **Astro**, **TypeScript** and **Supabase**. Takes payments through **Mercado Pago** and **Webpay**.
- **Mosaic Builder**: in-browser app that turns a photo into a brick mosaic, matching colour in **CIE Lab** against the supplier's real palette, and outputs the parts list.

SKILLS

Data & MLOps:
Python (pandas, Polars, numpy, scikit-learn, matplotlib), Advanced SQL, PostgreSQL, DuckDB, BigQuery, GCP Vertex AI, Docker, MLflow, Prefect, Playwright, Git.

BI & Analytics:
Tableau, Looker, QuickSight, Power BI, SAP IBP, SAP BW.

Web Development:
FastAPI, Next.js, TypeScript, Astro, Streamlit, Supabase, MCP servers.

Areas of Specialization:
Machine Learning (Regression, Classification, Clustering), Pricing, Segmentation, Process Automation, Pipeline Orchestration, Data Storytelling.

Languages:
Spanish (Native), English (C1 - Advanced, IELTS), French (A1 - Beginner).

Courses and Certifications:
Stanford Machine Learning Specialization (Ongoing), Google Machine Learning Bootcamp (2022), Web Programming (Harvard CS50), Scrum Foundation Professional Certificate, Python Programming (UC, DataCamp).

EDUCATION

B.S. & M.S. Industrial Engineering — IT Minor
Universidad Adolfo Ibáñez, Santiago, Chile 2014 – 2019

International Baccalaureate Diploma (IBDP)
Saint Gabriel's School, Santiago, Chile 2013